

ELEC5305 Project Feedback Two

Project Title

Attention-Based Temporal Localisation of Musical Instruments in Polyphonic Music

Student Information

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GitHub Project Site: <https://decarre.github.io/elec5305-project-550275044/>

GitHub Repository: <https://github.com/Decarre/elec5305-project-550275044>

Project Progress

This project studies whether a weakly supervised audio model can recognise instruments in polyphonic music and show when each instrument is active. The model learns from instrument labels for a whole audio clip, without using frame-level timing labels during training. I compare mean pooling with instrument-specific attention while keeping the audio features and frame encoder the same. The main question is whether attention can provide useful temporal information as well as clip-level instrument predictions.

Achievements to Date

- **Implementation:** I completed log-mel feature extraction, matched mean, max and AttentionMIC-style models, temporal post-processing, and a command-line workflow for validation, training and evaluation.
- **Fair evaluation:** The pipeline supports artist-separated data splits, stable track and clip lists, validation-only threshold selection, saved model checkpoints, and one final test evaluation with fixed thresholds.
- **Dataset audit:** The official MedleyDB v1/v2 metadata contains 196 tracks from 116 artists. In total, 117 tracks have matching time-based activation files. The two public sample songs also passed the audio and annotation checks.
- **Verification:** All 13 automated tests pass. They check model pooling, metrics, artist-separated data lists, saved checkpoints and temporal processing.

Preliminary Engineering Result

Model	Initial BCE	Final BCE	Micro-F1	Macro-F1
Mean pooling	0.4715	0.0357	0.8721	0.7717
Attention pooling	0.4509	0.0135	0.9977	0.9967

These values are engineering checks from 230 non-overlapping two-second clips taken from two public sample songs. The same clips were used for training and measurement. Therefore, the results confirm that the pipeline works, but they do not show performance on unseen music or prove that attention is better than mean pooling.

Current Limitation and Next Step

Access to the full MedleyDB audio has been requested and is still waiting for approval. If it is approved, complete artists will be separated into training, validation and test sets before the audio is divided into clips. Thresholds and temporal processing settings will be selected only on the validation set. The final model will then be tested once on unseen artists.

If MedleyDB access is delayed, the duplicate-free Slakh2100-redux split will be used for the main quantitative experiment. MedleyDB can then be used later as an external real-audio check. This backup plan keeps the same research question and prevents the dataset approval process from stopping the project.

Project Resources

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AI Use Statement

OpenAI Codex was used to help plan the workflow, draft and revise Python code, debug the local environment, prepare tests, and improve the English in this progress report. The author checked the code, test outputs, links and reported numbers against the repository files. The author made the final project decisions and remains responsible for the submitted work. AI was not used to create experimental data.